

Netflix Exploratory Data Analysis (EDA)

Netflix has emerged as one of the largest online streaming platforms, offering a wide variety of films and television shows to a global audience. This report presents an exploratory data analysis (EDA) of the Netflix dataset, examining its structure, country distribution, genres, ratings, and trends over time. The insights derived help us better understand Netflix's content strategy and audience engagement.

Dataset Overview

The dataset contains a total of 8,790 records with 10 key attributes, including show ID, type, title, director, country, release year, rating, duration, and genres. It features 6,126 movies and 2,664 TV shows, covering releases from 1925 to 2021.

Top Contributing Countries

Rank	Country	Count
1	United States	3240
2	India	1057
3	United Kingdom	638
4	Pakistan	421
5	Canada	271

Movies vs TV Shows

Movies dominate the Netflix catalog, accounting for approximately 70% of the total titles (6,126), while TV shows represent 30% (2,664). This indicates Netflix has historically prioritized film content, though TV shows are gaining significant traction.

Release Year Distribution

The majority of Netflix content has been released after 2010, with a peak observed during the years 2017–2019. This period highlights a phase of rapid expansion in Netflix's content library.

Top Genres

Some of the most frequent genres on Netflix include International Movies, Dramas, Comedies, Documentaries, and Action & Adventure. This reveals Netflix's focus on global storytelling, light entertainment, and dramatic narratives.

Ratings Distribution

The most common rating across Netflix titles is TV-MA, which targets mature audiences, followed by TV-14 and TV-PG. This pattern suggests Netflix caters predominantly to adult viewers, though it also includes family-friendly programming.

■ Movie Durations

The average duration of Netflix movies is around 100 minutes, with the majority falling between 80–120 minutes. This duration aligns with conventional movie lengths, although there are outliers at both extremes.

■ Correlation & Insights

Label encoding was applied to categorical attributes to enable correlation analysis. The resulting heatmap demonstrated only weak correlations, which is expected given the categorical nature of the data. Hence, Netflix data is best interpreted through categorical analysis such as counts and distributions rather than numerical correlations.

■ Conclusion

This analysis reveals several important insights: - The United States dominates Netflix's content, followed by India and the UK. - Movies represent the majority of titles, though TV shows are growing in importance. - Content production surged after 2010, peaking between 2017 and 2019. - Netflix's content leans towards drama, comedy, and international stories. - Ratings show a strong emphasis on mature audiences, reflecting global viewing trends. Overall, Netflix continues to expand its diverse library, balancing regional preferences with global demand.